The Direct and Indirect Effects of Genetics and Education

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Abstract

In-situ abstract, subject to change:

Social science genetics has shown genes matter for education and labour market outcomes, but has little credible evidence on the economic mechanisms beneath these findings. I use plausibly random deviation from parents as a natural experiment for the Education PolyGenic Index (Ed PGI) to estimate the causal effect of genes associated with years of education, with data from the UK Biobank. I then decompose the total genetic genetic effects into a direct genetic effect and indirect education pathway using a causal mediation framework that allows for selection-into-education, using multiple UK university openings in the 1960s as instruments for higher education attendance. The Ed PGI increases labour market earnings through higher education attendance, with direct genetic effects indistinguishable from zero — contradicting previous speculation that education-linked genes independently increase intelligence or other labour market traits. Education linked genes are more about returns to higher education, and less about direct genetic effects. These findings reframe the usual focus in social science genetics by shifting from asking "if genes matter" to "how genes matter" — not as deterministic forces, but as factors that influence human capital decisions within institutional constraints. By using causal methods that account for individual choice in education, and not than assuming genetic determinism, this research provides a richer understanding of how biological inheritance, human capital decisions, and institutional structures interact to shape labour market inequality.

Keywords: Education, Mediation, Polygenic indicies, Nature/nurture.

JEL Codes: D31, D91, I24, J24, Z00.

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> The difference of natural talents in different men, is, in reality, much less than we are aware of [...]. The difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise not so much from nature, as from habit, custom, and education.
>
> — *Smith (1776), The Wealth of Nations, Volume 1 p. 16.*

0.1 Writing Plan

Section map for this paper:

1. Intro, written last following `https://blogs.ubc.ca/khead/research/research-advice/formula`.
2. Data + biological context.
3. Section on the adjusted PGI randomisation design, clearly stating estimands
4. New results of genetic effects.
5. Conceptual framework for mediation effects, clearly stating estimands
6. Results of the mediation analysis.
7. Conclusion.

Note: a major concern for the Ed PGI previous papers is that they have little economic insight. This paper is framed as changing that, investigating the economic mechanisms underlying the Ed PGI causal effects, giving economic credence where before there was merely concern.

Papageorge & Thom (2020) document how the Ed PGI predicts earnings, even when controlling for education, and posit that the index contains genes that affect skills related to financial literacy.

Quote for the Mendelian independent first-stage, "Cannot choose who your parents are, and you also cannot choose how much you inherit from them."

1 Introduction

This paper proceeds as follows. Section 2 summarises the UKB data, and explains how the Ed PGI is constructed. Section 3 shows a causal research design, based on Mendelian independent inheritance, for estimating the causal effects of the Ed PGI; Section 4 shows

estimates of the causal effects of the Ed PGI, on education and labour market outcomes. Section 5 gives a conceptual framework for estimating the direct and indirect effects of genetics and education, in this observational setting; Section 6 gives the empirical results. Section 7 concludes.

2 Data and Genetic Context

2.1 UK Biobank (UKB)

The UK Biobank (UKB) is a large-scale prospective cohort study comprising approximately half a million members of the British public, recruited between 2006 and 2010. Letters were sent to a random sample of addresses in the National Health Service registry, and the final UKB data represent voluntary respondents.[1] Participants, aged 40-69 years at recruitment, underwent comprehensive baseline assessments, asked many questions for socioeconomic information either by interview, automated survey answers, or follow-up sessions. Key socioeconomic variables include educational attainment (measured in education qualifications), household income brackets, employment status, and industry of employment. See Sudlow et al. (2015) for a full description of the UKB organisation and data.

The UKB collected detailed information on educational qualifications, including completion of primary school, GCSEs (typically completed at age 16), A-Levels (completed at age 18), and completion of an higher education degree. I convert these qualifications into education years based on each participant's highest reported qualification, following the International Standard Classification of Education.[2] While the UKB collected household income data, this information was recorded only in categorical bins, limiting its utility for continuous analysis. Following Kweon et al. (2025), I construct an individual-level income measure using participants' 4-digit Standard Occupational Classification (SOC) codes. Specifically, I impute occupation-based wages by estimating hourly earnings for each SOC category using data from the British Annual Survey of Hours and Earnings, conditional on age, gender, and year. This approach provides a more granular and continuous measure of economic status than

[1] The UKB may not be representative of the entire UK population, as it only represents willing respondents. As such, the findings of this paper should only be taken as internally valid among the UKB respondents, and the sibling analysis sample described.

[2] As noted by Okbay et al. (2022), British vocational degrees inconsistently coded by the International Standard Classification of Education. I follow the literature by assigning education years as the age they reported leaving full-time education minus five, among those with a vocational degree as their highest qualification.

Table 1: Descriptive Statistics, UK Biobank (UKB).

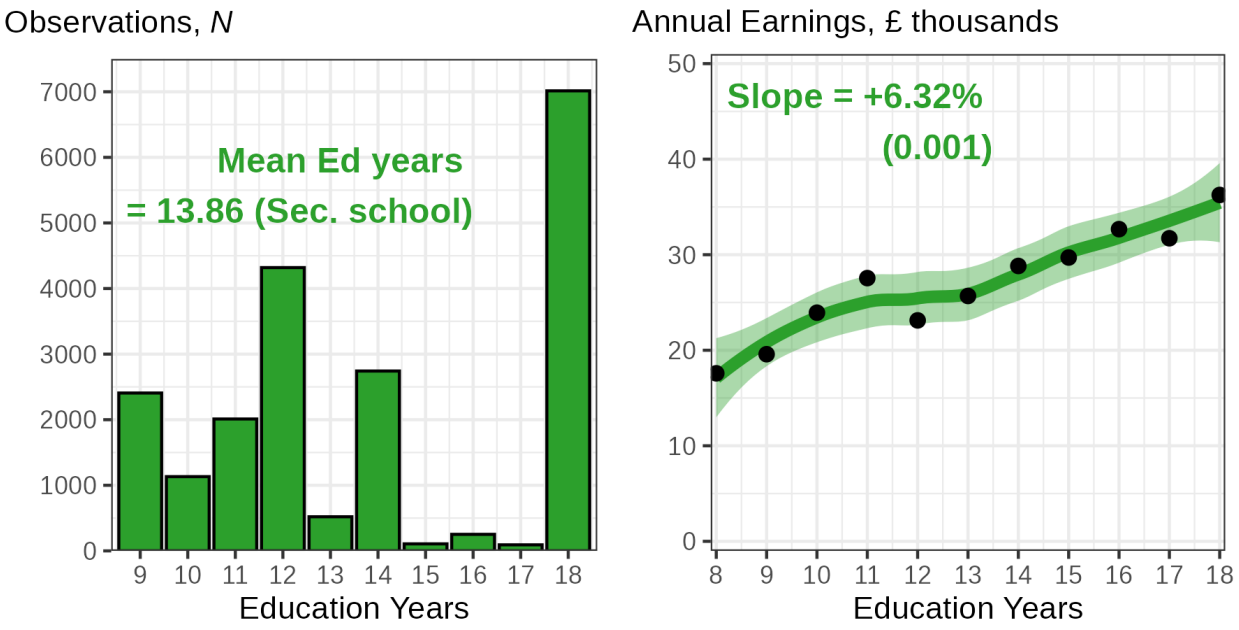
	Analysis sample		Entire sample	
	Mean	SD	Mean	SD
Demographics:				
Male	0.44	0.50	0.46	0.50
Age	55.04	7.05	56.55	8.09
Lives in city	0.86	0.35	0.85	0.35
Ethnicity = European	1.00	0.00	0.82	0.38
Any siblings also in UKB?	1.00	0.00	0.08	0.28
Count of siblings in UKB	1.09	0.32	0.09	0.32
Genetic Measures:				
Ed PGI	0.04	1.00	-0.00	1.00
Ed PGI, imputed parental mean	-0.02	0.75		
Other PGI: ADHD	-0.02	1.00	-0.00	1.00
Other PGI: Asthma	-0.01	1.00	0.00	1.00
Other PGI: Bipolar	-0.01	1.00	-0.00	1.00
Other PGI: BMI	-0.02	0.99	-0.00	1.00
Other PGI: Diabetes (T2)	-0.02	1.00	-0.00	1.00
Other PGI: Height	0.00	1.01	-0.00	1.00
Other PGI: Schizophrenia	-0.04	1.00	0.00	1.00
Education:				
Education years	13.84	3.36	13.72	3.48
Qualification, University degree	0.33	0.47	0.33	0.47
Qualification, A-Levels	0.46	0.50	0.45	0.50
Qualification, GCSEs	0.76	0.43	0.71	0.45
Qualification, Professional degree	0.32	0.47	0.29	0.45
Qualification, Vocational degree	0.23	0.42	0.19	0.39
Qualification, No official qualifications	0.13	0.34	0.17	0.38
Labour Market Outcomes:				
Occupation hourly wage, £	20.20	9.09	20.91	9.51
Occupation annual income, thousands £	28.67	18.85	30.40	19.94
Average hours worked, per week	34.73	12.58	35.31	12.70
Household income, $< £18k$	0.13	0.33	0.20	0.40
Household income, $£18 - 31k$	0.23	0.42	0.22	0.41
Household income, $£31 - 52k$	0.28	0.45	0.22	0.42
Household income, $£52 - 100k$	0.22	0.41	0.17	0.38
Household income, $£100k <$	0.05	0.22	0.05	0.21
Observations	24,755		496,204	

Note: This table shows the Mean and Standard Deviation (SD) of UKB variables, comparing the analysis subsample and the entire UKB sample. The analysis sample refers to individuals with at least one sibling in the UKB data, of European ancestry. All genetic variables are only non-missing among those who report European ancestry.

the categorical household income variable (see Appendix A.4 for details on the imputation procedure). I create the annual occupation income by multiplying this by the number of hours they reported working, times 52 weeks in the year. For an example, this procedure gives a female participant, age 58 in 2010 with SOC 2314 (i.e., a secondary education teacher by profession), an occupation hourly wage of £29.20 and an annual income of £68,476 for her 45 hours worked each week (adjusted to 2024 £GBP by CPI-H).

Figure 1: Distribution of UKB Education Measures.

(a) Education Years. **(b)** Education Years + Income.



Note: This figure shows the marginal distribution of education years in the UKB analysis sample, and the correlation between years of education and occupation coded income.

For the purposes of this study, the primary analysis sample is restricted to those with at least one family member, sibling or parent, also enrolled in the UKB, and non-missing data in occupation code plus education qualifications. The UKB does not personally identify participants, nor their family members, so I follow the genetic literature by identifying family clusters by measures of genetic similarity.[3] The presence of multiple family members within the dataset provides a unique opportunity to control for shared genetic factors that might otherwise bias estimates in standard cross-sectional analyses.

[3]The design of the UKB allows for a larger number of family clusters than would be expected from a random sample of the British population. This is because study enrolment letters were sent to households, and the household head (or to whomever the letter was addressed) could optionally enrol their family members in addition to themselves. Appendix A.2 describes the family cluster identification procedure in detail, which has been used in previous social science genetic studies (see e.g., Muslimova et al. 2025).

The analysis sample is also restricted to individuals who report to have primarily European ancestry. The genetic measures used in this paper have generally been constructing from primarily white, European descended populations, and have no predictive power out of sample, so are only relevant in this subsample.[4] The only notable differences between the analysis subsample and the entire UKB participant sample is that the analysis sample has more siblings, and is entirely European descended; no other variables are significantly different for the analysis subsample. Table 1 summarises the UKB data, for the analysis sample and among the entire UKB sample. Figure 1 summarises the correlation between education years and occupation coded income.

2.2 Education PolyGenic Index (Ed PGI)

UKB participants also provided saliva samples that were subsequently genotyped using high-density arrays containing over 800,000 genetic markers, producing standardised genetic data files for statistical analysis of the relationship between genetics and social outcomes.

While genetic data provide rich information about individual variation, the relationship between specific genes and social outcomes like education is extraordinarily complex. The biological pathways from DNA to observable traits remain largely opaque — the exact relationship between a single gene, its resulting protein (or its moderating role), and the effects of those proteins are only known for a small number of genes in the human genome. For example, a single gene (and its recessive variant) is known to lead to a protein which does not break down the by-products of alcohol digestion; individuals with this gene generally do not drink alcohol at all because of the simple limit on digestion (Millwood et al. 2023). Complex social behaviours, on the other hand, have no single genes with a known biological pathway, so cannot be summarised in a single gene, nor single biological pathway. Instead complex social behaviours are a result on thousands of genes, and their interactions with the environment.

Rather than attempting to trace all of these intricate biological mechanisms, researchers have developed PolyGenic Indices (PGIs) as dimensional reduction tools that summarise the statistical relationship between thousands of genetic variants and outcomes of interest.[5] This

[4] It is an unfortunate fact that systematic genetic measures, such as those considered in this paper, have only been constructed from Biobank samples where the vast majority are white and/or European descended; see Martin et al. (2017) for a further discussion. Note that this implies this paper has nothing to say about race; see Subsection 2.4.

[5] See Uffelmann et al. (2021) for a full explanation of how a GWAS works, and the approach to calculating PGIs.

aggregation approach is particularly valuable because individual genetic variants typically have tiny effects on complex traits like educational attainment — often explaining less than 0.01% of the variation. However, by combining information across hundreds of thousands of variants, each contributing a small effect, PGIs can capture meaningful portions of the genetic contribution to these outcomes. These indices are fundamentally correlational measures: they capture statistical associations between genetic variants and outcomes observed in large samples, not well understood biological pathways, and without a clear case for causal relationships. The weights used to construct PGIs reflect population-level correlations that may arise through various channels, including biological effects, gene-environment correlations, and population stratification.

The Education PGI (Ed PGI) is a summary measure that aggregates genetic information across thousands of genetic variants into a single index predictive of educational attainment. The construction of the Ed PGI relies on weights derived from a Genome-Wide Association Study (GWAS), which analyse the correlation between genetic variants and education years. In a GWAS, researchers test the correlation between each Single Nucleotide Polymorphism (SNP) — a one-letter change in the DNA sequence at a specific location — and the outcome of interest, in this case education years. I use the weights calculated in the GWAS conducted by Okbay et al. (2022), which used a sample of over 3 million individuals of European ancestry, primarily from 23&Me data.

The Ed PGI is defined as the weighted sum of whether individual i has the relevant SNP at each DNA location j,

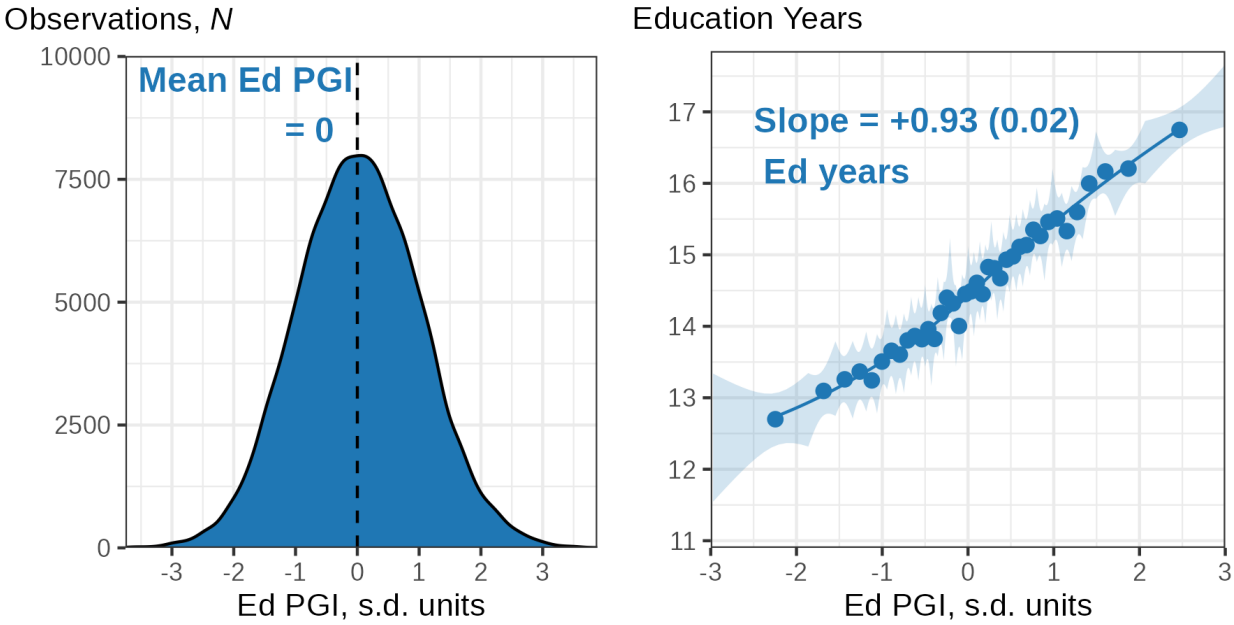
$$Z_i = \sum_{j=1}^{J} w_j x_{i,j}$$

where w_j are the Ed PGI weights (provided in the supplementary data to Okbay et al. 2022), and $x_{i,j}$ is an indicator of whether individual i has the relevant SNP at DNA location j.[6] The Okbay et al. (2022) GWAS found that over 8,000 individual SNPs were significant in predicting education years, after passing a battery of biostatistical tests and finding no evidence for significant interactions. The resulting Ed PGI is standardised to have mean zero and standard deviation one within the entire sample, making it interpretable as the number of standard deviations from the population mean. The Ed PGI captures approximately 12-15% of the variation in educational attainment, making it one of the most predictive PGIs for any complex behavioural trait to date.

[6]Every person has two copies of each chromosome, so that $x_{i,j}$ takes values 0, 1, 2 corresponding to how many copies of the SNP i has. The genetics literature has generally accepted that complex social behaviours are explained by additive effects (see Hill et al. 2008), and Okbay et al. (2022) found little evidence for dominance in the Ed PGI SNPs.

Figure 2: Distribution of Education PolyGenic Index (Ed PGI).

(a) Ed PGI. **(b)** Ed PGI + Education Years, Binscatter.



Note: This figure shows the marginal distribution of Ed PGI and education years in the UKB analysis sample. Ed PGI is normalised to have mean 0 and standard deviation 1 among the entire UKB sample; it is roughly normally distributed regardless of this normalisation.

The Ed PGI is highly correlated with education years, as it is constructed to be so. A one standard deviation increase in the Ed PGI is associated with 0.93 extra education years (standard error 0.02), with an R^2 value of 0.10 for the raw regression of education years on Ed PGI. It is roughly normally distributed, with standard deviation normalised to 1. It is not surprising that the Ed PGI is normally distributed; it is the weighted sum of thousands of binary variables, among a few thousand individuals, so we should expect the central limit theorem to apply. Additionally, it is not surprising that the Ed PGI is correlated with education years; it was literally constructed to be so. The main take-away is that the correlation is strong, and has meaningful predictive power for education years. If education had no relationship with genetic variants, then we may expect the correlation would be indistinguishable from zero.

The Ed PGI is not the only PGI that can be constructed. Following the same methodology, I construct additional PGIs for six health-related traits that may confound the relationship between genetics and socioeconomic outcomes: asthma, bipolar disorder, Body Mass Index (BMI), height, schizophrenia, and type 2 diabetes. Each PGI is calculated using the same approach as the Ed PGI — taking weights from a published GWAS, conducted on independent

populations, and computing weighted sums of the relevant genetic variants in the UKB data. These other PGIs serve as control variables in my analysis for two reasons. First, genetic variants often exhibit pleiotropy, meaning a single variant can influence multiple traits simultaneously. For instance, genetic variants associated with education might also affect other factors (e.g., mental health conditions), which could independently influence labour market outcomes. Second, systematic differences in gene frequencies across ancestry groups can create spurious correlations between any PGI and outcomes if health conditions vary across these groups for unrelated reasons.

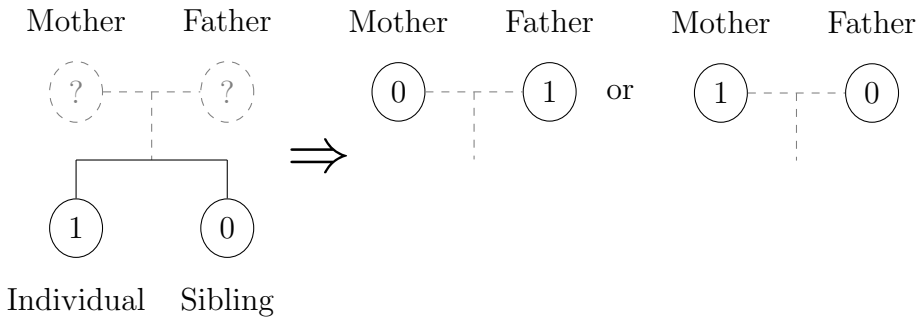
2.3 Imputing Parental PGIs from Siblings

It would be ideal if the UKB contained genetic information for the parents of every UKB participant. While there are 6,239 participants with a parent also included in the UKB, there are only 72 with both parents. To overcome this missing data problem, I use siblings' genetic data to impute the mean parental PGIs from the sample of 24,755 participants with at least one sibling also in the UKB — the sibling analysis sample described above.

I use the Young et al. (2022) procedure, to impute mean parental PGI values using genetic data on participants' siblings. To briefly summarise how this works, consider that each person inherits exactly half their DNA from their mother and half from their father. When we observe two siblings, we can partially reconstruct their parents' genetic information because siblings both inherit their DNA from parents. Consider observing two different samples from the same underlying distribution — the parents' genetics. At some genetic locations, siblings inherited the same variant from a parent (like both getting mum's brown-eye variant), whilst at others they inherited different variants (one got mum's brown-eye variant, the other her blue-eye variant). By comparing siblings' genetics across thousands of genetic variants, we can recover about 75% of their parents' combined genetic information — enough to construct a reasonably accurate parental PGI.

Figure 3 visualises the logic behind imputing parental SNP values based on an individual's, and their sibling(s)'s, genetic data. In the middle panel, the individual has genotype 1 and their sibling has genotype 0. Since the individual must have inherited their 1 allele from one parent and their "0" allele from the other, while the sibling inherited two 0 alleles, we can infer that one parent has genotype 0 (contributing 0 to both children) and the other has genotype 1 (contributing 1 to the individual and 0 to the sibling). The right panel shows an alternative configuration yielding the same sibling genotypes. When parental genotypes cannot be uniquely determined, the imputation uses the expected values based on observed

Figure 3: Imputing One SNP for Parents from Sibling Data.



Note: This figure illustrates the logic of imputing parental genotypes from sibling data at a single SNP. At each SNP, individuals have a genotype of 0, 1, or 2, representing the number of copies of a particular allele inherited from their parents.

alleles and population frequencies. By applying this logic across hundreds of thousands of SNPs and accounting for segments of DNA shared identically by descent between siblings, we can accurately reconstruct parental PGIs despite never observing the parents directly.

The imputation procedure performs well in the UKB analysis sample. Following the Young et al. (2022) validation approach, I verify the accuracy by examining the small subset of families where parental genotypes are actually observed. The imputed parental PGIs show a correlation of over [X] with true parental PGIs, and regression coefficients are statistically indistinguishable from 1, confirming the imputation is unbiased. Appendix A.3 describes the full technical detail behind the imputation procedure, and this validation exercise. This high accuracy means I can confidently use the imputed parental PGIs in place of directly observed parental PGI values.

2.4 Ethical Considerations

It is crucial to emphasise that the genetic measures in this study should not be misinterpreted as evidence of genetic determinism. Having a higher Ed PGI does not destine someone to higher education or income; it merely represents only a small statistical tendency, not fate. These tendencies are fundamentally shaped by social policies, institutional structures, and cultural contexts that vary across time and place. Just as eyesight correcting glasses can eliminate the practical consequences of genetic variation in eyesight, education and economic policies can modify or eliminate the consequences of genetic variation in traits relevant to socioeconomic outcomes.

Furthermore, this research provides no support whatsoever for ideas about "natural" social hierarchies or genetic justifications for inequality. In addition, genetic associations identified within one ancestry group cannot be meaningfully compared across groups due to different genetic architectures, environmental contexts, and historical experiences. Any claim using the measures consider in this paper for inference across racial groups, or different racial groups would be both scientifically invalid, and ethically deficient. The appropriate interpretation of these results is that they highlight how both social and genetic factors influence socioeconomic outcomes, which should inspire greater solidarity and support for policies that promote equality of opportunity rather than acceptance of inequality as natural or inevitable.

3 Random Variation in Genetic Inheritance

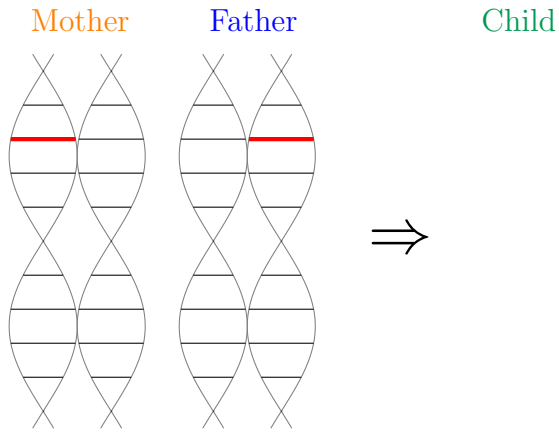
Genetic variants are not randomly assigned, which means that simple correlations between the Ed PGI and outcomes cannot be interpreted as causal relationships. Genes are inherited from parents, where children receive 23 chromosomes from each parent at conception — half the genes from their mother, half from father. This inheritance pattern creates systematic associations between an individual's genetics and their family environment, as children who inherit certain genetic variants also inherit the familial circumstances from parents carrying those variants. However, this inheritance pattern also involves independent random variation, which I use to instrument for changes in the Ed PGI and thus estimate causal effects.

3.1 Mendelian Independent Inheritance

I use variation away from parental values to estimate causal effects of having a higher versus lower Ed PGI. This approach leverages the fundamental randomness in genetic inheritance, first described by Mendel's laws (Mendel 1865).

Each parent has 0, 1, or 2 copies of each genetic variant — corresponding to presence on each of their chromosomes. When a father and mother's sex cells are formed, exactly one of the two copies is sorted into each reproductive cell, and is then inherited by their resulting child. A parent with one copy therefore passes it on half the time, whereas a parent with 0 or 2 copies transmits their genetic variant with certainty. At conception, the child receives one draw from the mother and one from the father, so the child's copy-count is simply the sum of two independent draws.

Figure 4: Mendelian Independent Inheritance of One Genetic Variant.



Note: This figure illustrates the process of Mendelian Independent Inheritance of genetic variants, for parents with each one copy of a variant (red) and the resulting possibilities for their child's inherited variants.

Write $x_{i,j}^{\text{Child}}$ for the number of copies of variant j that individual i has (0, 1, or 2), and $x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}$ for their respective parents $m(i), f(i)$. If the mother has two copies of variant j then they bequeath one copy to their child with certainty; if the father has one, then they bequeath one copy with probability $\frac{1}{2}$; if no copies, then they bequeath none with certainty.

$$\mathbb{E}\left[x_{i,j}^{\text{Child}} \,\middle|\, x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}\right] = \mathbb{E}\left[x_{i,j}^{\text{Child}} \,\middle|\, x_{f(i),j}^{\text{Father}}\right] + \begin{cases} 0, & \text{if } x_{m(i),j}^{\text{Mother}} = 0 \\ \frac{1}{2}, & \text{if } x_{m(i),j}^{\text{Mother}} = 1 \\ 1, & \text{if } x_{m(i),j}^{\text{Mother}} = 2. \end{cases}$$

The child then receives their other copy from their father, by the same process, giving the following representation in expectation,

$$\mathbb{E}\left[x_{i,j}^{\text{Child}} \,\middle|\, x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}\right] = \frac{1}{2}\left(x_{m(i),j}^{\text{Mother}} + x_{f(i),j}^{\text{Father}}\right).$$

After conception, the child has inherited genetic variants from both parents — the discrete realisation, and not the expected value. This result came about purely from random chance, conditional on at least one of parent having exactly one copy of the SNP, $x_{m(i),j}^{\text{Mother}} = 1$ and/or $x_{f(i),j}^{\text{Father}} = 1$.[7]

$$x_{i,j}^{\text{Random}} := x_{i,j}^{\text{Child}} - \mathbb{E}\left[x_{i,j}^{\text{Child}} \,\middle|\, x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}\right] \quad \longrightarrow \quad \text{Independently assigned.}$$

[7]Note both parents having 0 or 2 copies leaves $x_{i',j'}^{\text{Random}} = 0$, so that individual i' inherited no random variation in variant j'.

$x_{i,j}^{\text{Random}}$ is the random variation in presence of SNP at position j. From the possible values of the parents' variants, $x_{i,j}^{\text{Random}}$ equals zero if both parents have 0 or 2 copies, equals $\frac{1}{2}$ if the child actually inherited 1 copy when both parents had one copy each ($-\frac{1}{2}$ if they did not inherit), and 1 if they inherited 2 copies and the parents had one each (-1 if they inherited 0).

3.1.1 Mendelian Independent Inheritance in the Ed PGI

Moving from a single variant to the PGI simply means summing across the same random-segregation for all the variants. The Ed PGI is constructed by summing the 0, 1, or 2 copy indicators across the $J \approx 8,000$ different SNPs, each multiplied by the externally estimated weights $w_{i,j}$ that capture how strongly each SNP predicts education years. Because every variant copy enters the child's genome through an independent random draw (conditional on parental values), the child's Ed PGI is the weighted sum of thousands of such draws.

This means the Ed PGI can be decomposed into the component that was inherited as a result of a random draws from parent(s) who had 1 copy of the variant, and the certainty component from draws where parent(s) had 0 or 2 copies.

$$Z_i^{\text{Child}} = Z_i^{\text{Random}} + \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}\right],$$

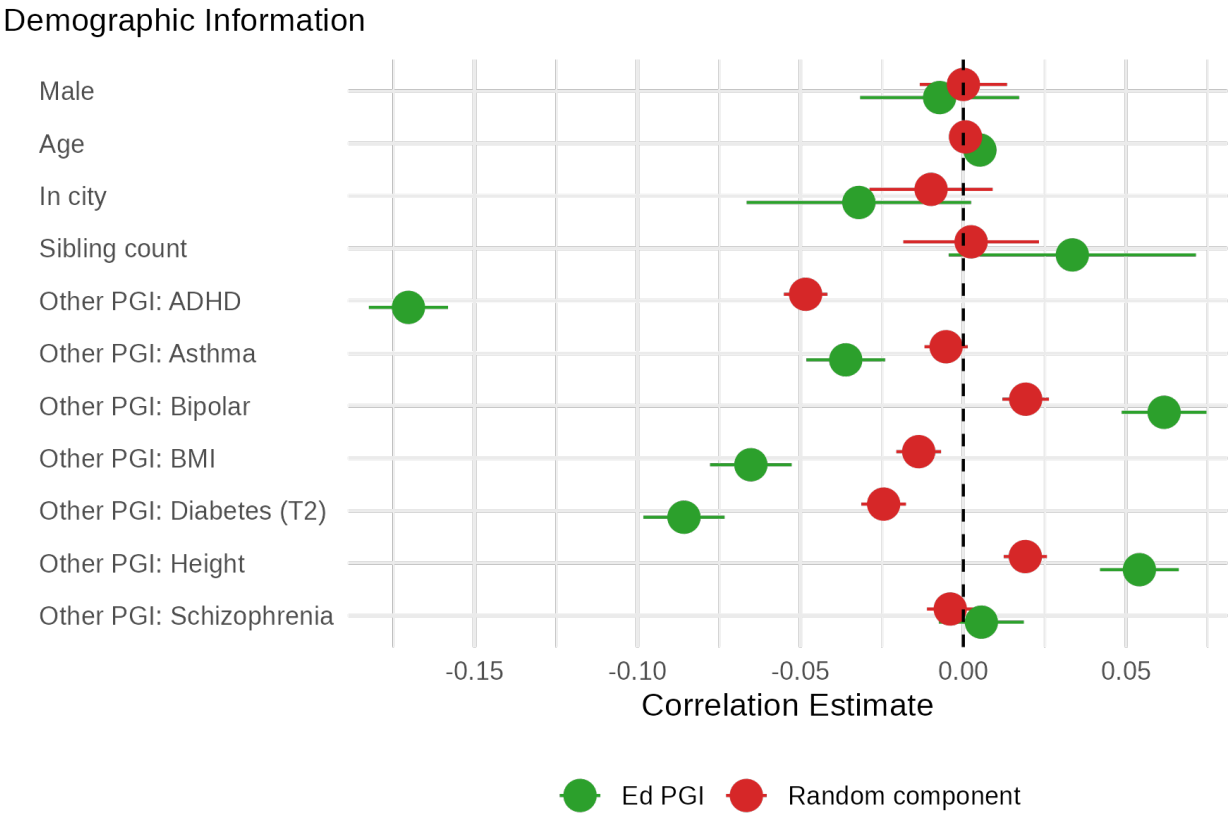
where $Z_i^{\text{Random}} = \sum_{j=1}^{J} w_j x_{i,j}^{\text{Random}}$ is the random component of the Ed PGI for individual i, and similarly for their parents' Ed PGI values, $Z_{m(i)}^{\text{Mother}}$ and $Z_{f(i)}^{\text{Father}}$. Z_i^{Random} is the random component because it was the result of Mendelian independent inheritance, and its only determining factor was having at least one parent with exactly one copy for relevant genetic variants — and nothing else.

Describe the random component. What is its distribution. It is not correlated with other demographic variables, and less so with the other PGIs. Figure 5 shows what the Ed PGI, and the calculated random component is correlated with.

3.1.2 Using Imputed Parental Ed PGI Values

The random component is a function of parental PGI values, which are not directly observed in the UKB data. I impute the mean of the parents value, $\frac{1}{2}\left(Z_{m(i),j}^{\text{Mother}} + Z_{f(i),j}^{\text{Father}}\right)$, using the genetic data from individuals who have siblings in the UKB. This is the method proposed by Young et al. (2022), who showed that the genotypes of a pair of siblings contain almost

Figure 5: Correlation for Ed PGI, or Random Component, and Demographic Variables.



Note: Multivariate regression estimates for the linear model $Z_i = \boldsymbol{\phi}' \boldsymbol{X}_i + \eta_i$, for each Z_i^{Child} and Z_i^{Random}, where $\boldsymbol{X}_i$ is the vector demographic variables described by the labels on the y-axis.

all the information needed. For any given SNP the two children collectively reveal between two and four of the four parental variants, depending on how many variants the siblings share identically-by-descent (IBD). Averaging over the three possible IBD states, the analyst observed 3 out of 4 variants in expectation; the final unobserved variant is imputed with its population frequency. Taking weighted sums across the thousands of variants therefore yields imputed parental PGIs that are extremely highly correlated with the true parental values — see Subsection 2.3.

Substituting these imputed values in place of the missing parental values does not bias the downstream regression; Young et al. (2022) the resulting estimators of the effects remain unbiased and consistent, and that the usual robust variance formula is valid. The chief cost is a modest loss of precision: depending on genetic data availability, and the proportion of siblings' genetic variants that end up being identical-by-descent, the effective sample size is one-half to one-sixth of what it would be if both parents were actually observed. Even so,

this represents a substantial gain over the classical sibling-difference design.

The random genetic variation leveraged in this analysis fundamentally arises from siblings who inherit different sets of variants from their parents. An individual and their sibling may inherit Ed PGI values far apart, from randomly one inheriting more copies of relevant variants from their than their sibling. The resulting variation provides a natural experiment, offering a source of quasi-experimental variation. In contrast, sibling pairs whose inherited Ed PGI values close to each other (or equal) produce minimal informative variation. Thus, the effective random variation in this framework predominantly stems from sibling pairs whose genetic inheritance diverges notably.

Consequently, the derived random genetic component is inherently a noisy measure of true Mendelian random variation. Its precision depends on how genetically distinct each sibling pair is: sibling pairs who differ widely provide clearer signals, while genetically similar pairs introduce greater measurement noise. For estimation, an Ordinary Least Squares (OLS) regression estimate will implicitly weight observations by variance (Angrist 1998), meaning sibling pairs with larger genetic differences carry greater influence on estimated effects. The resulting estimates disproportionately reflect families where sibling genetic inheritance diverged substantially, potentially affecting generalisability if sibling genetic divergence correlates systematically with family characteristics.

3.2 Measurement Error in the Ed PGI

An important econometric consideration is that the Ed PGI weights, $w_{j'}$, are themselves statistical estimates rather than true population parameters. Although the Okbay et al. (2022) weights are estimates from an exceptionally large sample of over 3 million individuals, they remain subject to finite sample sampling variation and estimation error; the sample size if large, but the number of estimated parameters is similarly large ($w_{j'}$ for around 8,000 different values), making estimation error a sufficient concern. This measurement error in the weights propagates through the observed Ed PGI, resulting in classical attenuation bias when examining correlations between the PGI and outcomes of interest.

To address this measurement error problem, I follow van Kippersluis et al. (2023) and construct a secondary Ed PGI using alternative weights, also provided by Okbay et al. (2022), that were estimated exclusively from the UK Biobank non-sibling subsample. This Secondary Ed PGI is noisier, and has a lower correlation with education years than the Primary Ed PGI described above, because it was calculated on a smaller sample size — and so has higher

measurement error. However, this measurement error comes from estimating on a different sample to the Primary Ed PGI, so their respective measurement errors are uncorrelated. Thus, I use the Ed PGI random component, calculated with weights from the secondary Ed PGI, as an instrument for the child's Ed PGI, calcuated with weights from the primary Ed PGI.

This approach trades off bias for higher variance, where the measurement error was causing attenuation bias, but introducing a secondary Ed PGI increases the variance from implementing the IV procedure. To alleviate increasing variance, I estimate the approach using Obviously Related Instrumental Variables (ORIV, Gillen et al. 2019). ORIV works as follows: calculate the IV results with the secondary Ed PGI instrumenting for the primary Ed PGI ($1 \rightarrow 2$), and similarly calculate the IV results with the primary Ed PGI instrumenting for the secondary Ed PGI ($2 \rightarrow 1$). Then stack together these results, and adjust the standard errors for their stacked combination. This approach uses the primary and secondary Ed PGIs as instruments for each other, overcoming attenuation bias in the most efficient way possible by combining IV first-stages. To my knowledge, this is the first use of this method to estimate causal PGI effects (while also using the random component and/or controlling for parental values),[8] and so updates the literature regarding the size of PGI causal effects.

3.3 Genetic First-Stage Effects

The Ed PGI random component, Z_i^{Random}, serves as a formula instrument for an individuals' Ed PGI, Z_i^{Child}. Subsubsection 3.1.1 derived the random component, and its formula relating it to an individual's PGI.

I estimate the genetic first-stage as follows,

$$Z_i^{\text{Child}} = \gamma + \theta Z_i^{\text{Random}} + \boldsymbol{\zeta}' \boldsymbol{X}_i + \eta_i, \tag{1}$$

where θ is the first-stage average parameter, and $\boldsymbol{X}_i$ is a vector of controls including parental imputed PGI levels, gender, age, sibling count, in city indicator, and the other PGIs (i.e., all demographic variables in Figure 5). Since Z_i^{Random} serves is a formula instrument, it must be re-centred with respect to the control variables to produce valid causal effects (Borusyak & Hull 2023). I do this by calculating $\mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}\right]$ with interactions for the imputed parental PGI values and demographic variables.

[8] Previous use of ORIV in the context of PGIs uses the approach to purge attenuation bias in correlational estimates, not causal effects (as in this paper).

Table 2: First-stage Genetic Estimates, $Z_i^{\text{Random}} \rightarrow Z_i^{\text{Child}}$ Independently and with ORIV.

	Outcome:					
	Primary Ed PGI, Z_i^{Child}			Secondary Ed PGI, Z_i^{Child}		
	(1)	(2)	(3)	(4)	(5)	(6)
Primary Ed PGI, Z_i^{Random}	1			0.204		
	(0.001)			(0.012)		
Secondary Ed PGI, Z_i^{Random}		0.158			1	
		(0.011)			(0.001)	
Combined, ORIV			0.235			0.262
			(0.003)			(0.003)
F-statistics		207	6434	102		7147
R^2	0.998	0.091		0.051	0.994	
Observations	24755	24755	24755	24755	24755	24755

Note: Shows estimates of the genetic first-stage, (1), using the random component as an instrument for the Ed PGI. The first three columns using the primary Ed PGI as the outcome, while the next use the secondary Ed PGI. The row marked ORIV uses the combined estimates of the primary and secondary Ed PGI random component on the Ed PGI, estimated using ORIV.

The Ed PGI random component is a strong instrument for the Ed PGI, even when using the primary as an instrument for the secondary ($1 \rightarrow 2$, and visa versa). Table 2 summarises estimates of the genetic first-stage defined in equation (1). Columns (1) and (4) show mechanical results, equal to one by construction, as the random component is calculated from the primary and secondary Ed PGIs respectively. Columns (2) and (3) show that the random component is a strong instrument for the Ed PGI, calculated using weights from the other sample (primary or secondary), confirming that the random component from one PGI retains predictive power for the other, independently calculated PGIs. The fact that these cross-PGI coefficients are not equal to one indicates substantial measurement error in both PGIs, which would lead to attenuation bias in standard IV estimation. The ORIV estimates (columns 3 and 6) explicitly addresses this measurement error by combining information from both primary and secondary first-stages. Although ORIV estimates recover unbiased coefficients, this approach necessarily comes at the expense of greater variance.

4 Genetic Effects

Section to show the results of the Ed PGI → Education, labour earnings. Clearly define the estimands, and then the estimating equations.

Write D_i for education years, and $D_i(z')$ the potential outcome for Ed PGI value z'.

Write Y_i for resulting labour market income, and $Y_i(z', d')$ the associated potential outcome for Ed PGI value z' and education years d'.

Assumptions,

$$Z_i^{\text{Random}} \perp\!\!\!\perp D_i(.), Y_i(.,.)$$

and

$$Z_i^{\text{Random}} \text{ is a good instrument for } Z_i^{\text{Child}}$$

This gives identification for the effects of the Ed PGI on education years and labour market income,

$$\frac{\partial}{\partial z'}\mathbb{E}\left[D_i(z')\right], \text{ and } \frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(z', D_i(z'))\right].$$

Note that this is standard; the random component is strong instrument by design (a formula instrument), and everyone is a complier — though the OLS estimate gives more weights to those with higher Z_i^{Random} values.

4.1 Genetic Results

This will require using the random components as obviously related instruments for the other PGIs, respectively.

These results will likely be larger than in other papers, because of the measurement error reducing precision in previous work.

Figure 6: TODO: Make this show the binscatter of Ed PGI → Ed years, and income. Regular colour line for the raw correlation, and yellow line for the random component.

(a) Earnings. **(b)** Ed PGI + Earnings.

Note: This figure shows the marginal distribution of Ed PGI and education years among UKB household heads, in the analysis sample defined above. Ed PGI is normalised within-in sample by the UKB to have mean 0, and standard deviation 1; the distribution shape is unaffected by the normalisation — i.e., it is roughly normally distributed regardless. 3.4% of the analysis sample reports education years less than 10 years (not included in panel b).

TODO: a LaTeX table summarising the results, too.

5 Direct and Indirect Effects

This section gives the conceptual framework for direct and indirect effects.

Take the notation from beforehand, and clearly define the mediation estimands, which are average causal response functions in the mediation settings.

Why is this interesting and different from what has preceded?

5.1 Independent Variation in the Education Channel

Reference my other paper, showing that an informal mechanism analysis is appropriate here. Needs independent variation in the education channel to credibly estimate the mechanisms behind the total genetic effects.

Explain this with a simple Roy model.

Note that this now allows for a choice channel, when previously it had not.

6 Direct and Indirect Effects Results

7 Summary and Concluding Remarks

Conclusion goes here.

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A Supplementary Appendix

This project used computational tools which are fully open-source. Any comments or suggestions may be sent to me at `seh325@cornell.edu`, or raised as an issue on the Github project.

A number of statistical packages, for the R language (R Core Team 2023), made the empirical analysis for this paper possible.

- *Tidyverse* (Wickham et al. 2019) collected tools for data analysis in the R language.

A.1 Occupational Imputed Wages and Income

Describe here the Kweon Koellinger imputation procedure. Reference their supplementary data, and thanks for additional guidance on it.

A.2 Identifying Family Clusters by Genetic Similarity

Describe here.

A.3 Validation of the Young et al. (2022) Imputation

A.4 Two PGIs as Obviously Related Instruments